Governance-Aware Optimization (GAO)

Lawful Gradient Geometry for Controlled Parameter Evolution

Optimization becomes geometric shaping of update admissibility: updates are projected, bounded, or scaled according to declared invariants and protected subspaces.

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Abstract

Modern optimizers such as SGD, Adam, and AdamW are designed to accelerate convergence and stabilize learning under noisy gradients. They are not designed to encode authority, preserve semantic invariants, or regulate the legitimacy of parameter transformation. In contemporary large-scale training systems, this creates a structural asymmetry: governance may constrain deployment and evaluation, but optimization itself remains an ambient authority over parameter space.

This paper introduces **Governance-Aware Optimization (GAO)**, a framework for integrating explicit law into the update rule itself. Rather than treating governance solely as an external admissibility predicate, GAO defines a governance transform over proposed updates, shaping gradient-based transitions before they reach canonical state. Optimization becomes **geometric shaping of update admissibility**: updates are projected, bounded, or scaled according to declared invariants and protected subspaces.

GAO is not a new optimizer in the conventional sense. It is a **transformation layer** that composes with existing optimizers. GAO assumes a declared **Training-Law artifact** specifying protected subspaces, numeric bounds, invariants, drift monitors, and escalation policies. By distinguishing numeric descent from admissible motion, GAO preserves optimizer mechanics while embedding constraint, refusal, and structural stability directly into parameter evolution.

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1. Motivation

Training pipelines increasingly operate in environments where parameter evolution has institutional, safety, and operational consequences. Fine-tuning may

degrade protected behaviors. Hyperparameter drift may silently alter deployment characteristics. Safety heads or calibration layers may be overwritten in pursuit of narrow task performance.

Governed Training Regimes (GTR) introduce an admissibility membrane around optimization. GAO asks a deeper question:

Can governance influence the geometry of optimization itself?

Traditional constrained optimization embeds penalties inside the objective. GAO instead treats governance as a transformation over proposed updates. The distinction is structural:

- Constrained optimization modifies what is minimized
- Governance-aware optimization modifies what is permitted to change

GAO is compatible with GTR's episode-level gating but operates at finer granularity: **GTR asks whether a checkpoint may be promoted; GAO asks whether each individual update is admissible.**

GAO operates in differentiable, gradient-based training regimes. It assumes parameter updates are computed through conventional optimizers and inserts governance between update proposal and state commitment.

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2. Formalization of Admissibility

Let parameter space be $\mathbb{R}^n$.

Define the **admissible region** of update vectors:

$$\mathcal{A}(\theta_t, \mathcal{L}) \subseteq \mathbb{R}^n$$

where $\mathcal{L}$ is declared Training-Law.

A conventional optimizer proposes:

$$\Delta_t^{\text{prop}} = \mathcal{O}(\theta_t, B_t, H)$$

GAO defines a governance transform:

$$\Delta_t^{\text{gov}} = \mathbf{G}(\theta_t, \Delta_t^{\text{prop}}; \mathcal{L}) \in \mathcal{A}(\theta_t, \mathcal{L})$$

State evolution becomes:

$$\theta_{t+1} = \theta_t + \Delta_t^{\text{gov}}$$

The optimizer proposes. Governance restricts. Commitment follows.

Admissibility is geometric: it is the region of update vectors permitted under declared law. The admissible region ($\mathcal{A}(\theta_t, \mathcal{L})$) may be **convex** (simple projection), **manifold-constrained** (tangent projection), or **non-convex**.

For non-convex $\mathcal{A}$, GAO may require approximate projection methods (e.g., Dykstra-style alternating projections for intersecting constraints) or refusal when no admissible direction exists. The non-linear nature of $\mathbf{G}$ may induce non-convex effective admissibility behavior, even when individual governance components are convex.

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3. Objective-Space vs Update-Space Constraint

Penalty-based methods modify the loss:

$$L'(\theta) = L(\theta) + \lambda C(\theta)$$

This alters the gradient:

$$\nabla L' = \nabla L + \lambda \nabla C$$

Constraint is encoded into objective descent.

GAO leaves the objective and raw gradient untouched. The optimizer computes its internal state (e.g., momentum, adaptive scaling) using conventional rules. Governance operates after the optimizer proposes an update.

Thus:

- Penalty methods alter descent direction.
- GAO alters update admissibility.

Numeric optimization and admissibility control occupy distinct layers.

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4. The Governance Transform

Training-Law ($\mathcal{L}$) may include:

- Protected parameter masks ($\mathcal{P}$)
- Numeric bounds ($\mathcal{B}$)
- Declared invariants ($\mathcal{I}$)
- Drift monitors ($\mathcal{S}$)
- Escalation policies ($\mathcal{E}$)

($\mathcal{L}$) is an inspectable governance artifact. It may be versioned or externally adjudicated, but within GAO it defines the admissible update region ($\mathcal{A}$).

GAO projections relate to constrained optimization and projection methods in classical literature (Bertsekas, 1999; Boyd & Vandenberghe, 2004), but uniquely declare protection via Training-Law and place governance **post-optimizer, pre-commit**.

4.1 Protected Subspace Projection

Partition parameters into:

- Protected subspace (P)
- Governed subspace (G)
- Exploratory subspace (X)

Let (M_P, M_G, M_X) be mutually exclusive masks such that:

$$M_P + M_G + M_X = 1$$

Let ($\beta_G < \beta_X$) be per-subspace bounds declared in ($\mathcal{L}$).

Define:

$$\Delta_t^{\text{gov}} = 0 \cdot (M_P \odot \Delta_t^{\text{prop}}) + \text{clip}(M_G \odot \Delta_t^{\text{prop}}, -\beta_G, \beta_G) + \text{clip}(M_X \odot \Delta_t^{\text{prop}}, -\beta_X, \beta_X)$$

Protected parameters do not move. Governed parameters move under stricter bounds. Exploratory parameters retain wider authority.

This is structured authority partitioning, not aggregate clipping (Han et al., 2015; Nocedal & Wright, 2006).

text Example masks: (M_P = one-hot(`safety_head`)), (M_G = one-hot(`task_layer`)), (M_X = one-hot(`adapter`)).

text Example: (M_P) zeros safety classifier weights, (M_G) clips task-specific layers to 10% norm change, (M_X) allows full freedom for new adapters.

4.2 Invariant-Preserving Projection

Let an invariant ($I(\theta)$) be declared under law. GAO's goal is to preserve the invariant locally by remaining on its level set.

Project the proposed update onto the tangent space of the invariant manifold by removing its component along the normal direction ($\nabla I(\theta_t)$):

$$\Delta_t^{\text{gov}} = \Delta_t^{\text{prop}} - \frac{\langle \Delta_t^{\text{prop}}, \nabla I(\theta_t) \rangle}{|\nabla I(\theta_t)|^2 + \epsilon} \nabla I(\theta_t)$$

By construction:

$$\langle \Delta_t^{\text{gov}}, \nabla I(\theta_t) \rangle \approx 0$$

and therefore:

$$I(\theta_t + \Delta_t^{\text{gov}}) \approx I(\theta_t)$$

For multiple invariants (I_k), one may apply sequential or alternating projections. When invariant manifolds intersect non-transversally, sequential projection may fail to converge to a common admissible direction. In such cases, GAO may require approximate projection, relaxation, or refusal.

This follows classical tangent-space projection geometry (Bertsekas, 1999; Nocedal & Wright, 2006).

4.3 Drift-Aware Authority Scaling

Let drift measure (d_t) derive from KL divergence, calibration shift, or other declared metrics.

$$\Delta_t^{\text{scaled}} = \phi(d_t) \cdot \Delta_t^{\text{prop}}$$

Concrete choices:

- Linear: $(\phi(d_t) = \max(0, 1 - \gamma d_t))$
- Exponential: $(\phi(d_t) = e^{-\lambda d_t})$

To avoid oscillatory behavior, (ϕ) should be differentiable and Lipschitz continuous with constant $(L < 1)$.

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5. Ordering and Optimizer Interaction

For adaptive optimizers (e.g., AdamW), internal state is computed from raw gradients. GAO operates on the proposed update after these computations.

Recommended ordering:

1. Compute gradients
2. Construct optimizer update (including weight decay)
3. Apply GAO transform
4. Apply learning rate scheduling consistently

Repeated projections can alter effective momentum and step direction. Monitoring constraint activity is essential.

5.1 Governance Contract for ($\mathcal{L}$)

Governance contracts must be:

1. Explicit and inspectable
2. Versioned and hash-bound to training episodes to ensure auditability and prevent tampering
3. Declarative of protected subspaces and invariants
4. Auditable for drift metrics

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6. Relationship to Continual Learning

Continual learning methods infer parameter importance (e.g., via Fisher information). GAO instead uses declared invariants and masks.

EWC-style protections are implicit and difficult to audit. GAO protections are explicit and inspectable.

GAO relates to constrained optimization and continual-learning regularization (Bertsekas, 1999; Kirkpatrick et al., 2017; Zenke et al., 2017; Mallya & Lazebnik, 2018), but uniquely defines admissibility via declared law rather than inferred importance.

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7. Computational Considerations

Overhead includes:

- O(n) masking and clipping
- Optional invariant gradient computation
- Drift metric evaluation

Sparse tensors and precomputation can reduce overhead.

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8. Law Mutability

Training-Law ($\mathcal{L}$) may evolve.

Staged transition:

$$\mathcal{L}_t = (1 - \alpha_t)\mathcal{L}_{\text{old}} + \alpha_t \mathcal{L}_{\text{new}}$$

Constraint activity should be monitored:

$$\rho_t = |\Delta_t^{\text{gov}}|/|\Delta_t^{\text{prop}}|$$

Persistently low (ρ_t) indicates overconstraining and should trigger law review.

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9. Failure Modes

- Overconstraint -> increased convergence time, persistently low step ratio (ρ_t)
- Underconstraint -> invariant violation frequency
- Metric gaming -> adversarial satisfaction of declared metrics while violating intent (Goodhart's Law)
- False confidence -> "governance theater" if ($\mathcal{L}$) is poorly specified
- Interpretability issues -> projected updates may be difficult to diagnose without explicit logging of constraint activity and projection deltas

Internal governance reduces violation likelihood but does not replace external admissibility checks.

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10. Admissibility Preservation

If ($\mathcal{A}$) is convex and ($\mathbf{G}$) is the exact orthogonal projection onto ($\mathcal{A}$), then:

$$\Delta_t^{\text{gov}} \in \mathcal{A}$$

For non-convex ($\mathcal{A}$), GAO may require approximate projection methods (including alternating projection schemes such as Dykstra-style methods for intersecting constraints) or **refusal** when no admissible direction exists.

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11. Limitations

GAO cannot:

- Detect violations not encoded in ($\mathcal{L}$)
- Compensate for poorly specified invariants
- Replace evaluation-time governance
- Guarantee global safety properties
- Ensure feasibility when invariants conflict or admit empty intersections

GAO introduces new hyperparameters (e.g., (β_G), (β_X), and drift scaling parameters) that require tuning. Multiple governance transforms may not commute; order matters and can limit composability.

It shapes updates; it does not adjudicate legitimacy independently.

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12. Micro-Examples

```
## Composite GAO
delta_prop = optimizer.step()
delta_gov = GAO.subspace_projection(delta_prop, L)
delta_gov = GAO.invariant_projection(delta_gov, L)
delta_gov = GAO.drift_scale(delta_gov, L)
theta_t += delta_gov

## Drift scaling example
d_t = KL_divergence(model_logits, reference_logits)
alpha_t_gov = alpha_t * exp(-lambda_ * d_t)
delta_gov = alpha_t_gov * delta_prop
```

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13. Broader Implications

GAO reframes optimization as authority-constrained geometry.

Parameter evolution is movement within a law-defined admissible region. By separating numeric descent from update admissibility control, GAO introduces a new systems design axis:

Optimization dynamics vs Update admissibility geometry

GAO enables transparent governance: ($\mathcal{L}$) is inspectable, versioned, and auditable. GAO also supports dynamic governance: ($\mathcal{L}$) can evolve during training, but changes must be staged and monitored.

See companion work on **Governed Training Regimes (GTR)** for episode-level gating compatible with GAO.

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14. Conclusion

Optimizers were built for speed and stability, not legitimacy.

Governance-Aware Optimization introduces law into the geometry of gradient descent. Updates are proposed by numeric rules but shaped by declared admissibility constraints before they alter state.

GAO composes with existing optimizers. It does not redefine descent; it defines admissible motion.

GAO is not a silver bullet. It requires careful specification of ($\mathcal{L}$), explicit monitoring of governance effects, and external governance to preserve legitimacy.

By partitioning parameter subspaces and shaping update admissibility, GAO enables structured exploration while preserving critical invariants.

Corpus-Wide Reference Positioning

This paper argues that modern optimizers such as SGD, Adam, and AdamW are designed to accelerate convergence and stabilize learning under noisy gradients. Its nearest external comparison fields are governance, markets, formal. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. Optimization can make a chosen objective efficient while preserving a bad objective, missing distributional harms, strategic behavior, or inadmissible states. Formal assurance remains relative to the encoded model, trusted base, and refinement chain; proof does not remove specification error or unmodeled environmental behavior. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, 1990. DOI: 10.1017/CBO9780511807763. https://doi.org/10.1017/CBO9780511807763 **Relationship:** Predecessor institutional account of rule-governed collective action and polycentric governance.

[**E2**] Wil Waluchow and Dimitrios Kyritsis. "Constitutionalism." *Stanford Encyclopedia of Philosophy*, substantive revision 2023. https://plato.stanford.edu/entries/constitutionalism/ **Relationship:** Adjacent legal-philosophical account of limited authority; included to expose differences between political and computational constitutionalism.

[**E3**] Lawrence Lessig. *Code: Version 2.0*. Basic Books, 2006. https://codev2.cc/ **Relationship:** Predecessor argument that software architecture regulates behavior; the paper's contribution may be to make such regulation inspectable and revisable.

[**E4**] OECD. *Recommendation of the Council on Artificial Intelligence*. OECD/LEGAL/0449, adopted 2019 and updated 2024. legalinstruments.oecd.org source **Relationship:** Adjacent policy standard for accountability, traceability, transparency, safety, and human-centered values.

[**E5**] Harry Markowitz. "Portfolio Selection." *The Journal of Finance* 7(1), 1952, pp. 77-91. DOI: 10.1111/j.1540-6261.1952.tb01525.x. https://doi.org/10.1111/j.1540-6261.1952.tb01525.x **Relationship:** Predecessor risk-return optimization model; included to distinguish scalar optimization from governance-aware admissibility.

[**E6**] U.S. Department of the Treasury. *Tax Expenditures*. Office of Tax Analysis. home.treasury.gov source **Relationship:** Primary official baseline for the definition and estimation limits of U.S. federal tax expenditures.

[**E7**] George C. Necula. "Proof-Carrying Code." *POPL '97*, 1997, pp. 106-119. DOI: 10.1145/263699.263712. https://doi.org/10.1145/263699.263712 **Relationship:** Predecessor for binding machine-checkable evidence to executable

artifacts; the present paper may demand broader semantic or constitutional obligations.

Ableman corpus

[**A1**] Adam Ableman. *Ableman Constitutional Governance.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-79353AF163A4B0F1;
source SHA-256
FB0D9B2D3D54168FFE70B24308D1B2766D6D663ED24502D475E975939BE97C8B.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Constitutional Theory.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-D2D97C4042FD281F;
source SHA-256
FACC256EDDE5A7D7EA18133CFC9C060CFB8C3811B0221E1969E62347F2BB69CF.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Governance Engineering.* Ableman Corpus current edition; work identity PAPER.PUB.18749870;
source SHA-256
372169F315D192733856B1F4A3F1038BBC73F57CF3ED9ACABAE72B965A69C8D0.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Universal Governance Kernel (UGK).* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-1F9BFB2BF8370226;
source SHA-256
AA5106EA71E5DF120E141435C280DFC532715CA5DE0C75B8862BB7784ECCFACD.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Legitimacy Without Self-Ratification.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-EF2382F9397335AC;
source SHA-256
0186A39C808F8FDDB5C8F42384DF1A26678928434BD50A0EAE8D4947390F68A2.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Constitutional Markets.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-1D49CCBB6E47B403;
source SHA-256

BB97EAC6F1861C2ED17DC0E0B75D47B04716E7FF3E3AC2A9B2F6CABCC67A5250.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *ACG for Quants*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-8A19CB4C1BD55BCA;
source SHA-256
2FEA83087B8DA79472BBA449C9D5B252B6DBA67C2EAF0DF41C206B1B0EEEF254.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
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